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特邀报告摘要

**Abstracts of Invited
Speeches**



Alain Polguère
Université de Lorraine

Standard Lexical Functions, from Top to Bottom

Abstract: Standard lexical functions have been used for decades in Explanatory Combinatorial Lexicography to model recurrent semantic and combinatorial relations between lexical units of various natural languages. They are a powerful descriptive tool for at least two reasons. Firstly, they truly tie the language together, being at the interface between lexicon and grammar, on the one hand, and semantics, syntax and morphology, on the other hand. Secondly, they are organized in a formal, coherent system, which makes them particularly exploitable in computational lexicography and its applications. In spite of this, lexical functions have remained under-exploited by lexicographers and language specialists, largely because they are based on a complex system of theoretical notions whose mastering is a prerequisite to the acquisition of the very notion of lexical function. The aim of this talk is to try to break the learning barrier and give some essential keys to the use of lexical functions in lexicography and beyond (e.g., in language teaching).



Ana Frankenberg
University of Surrey

Who will Carry on Using Dictionaries in the Age of AI?

Abstract: Dictionary use studies have shown time and again that dictionaries tend to be vastly underused, with publishers generally underestimating people's metalinguistic skills. The call for dictionary training only goes to show that lexicography researchers have long been aware of the need to address the problem.

Despite the dramatic usability improvements that have taken place with the development of electronic dictionaries over the past decades, the competition posed by the recent popularisation of AI tools has brought into question the very existence of conventional dictionaries.

Although studies have demonstrated that good dictionaries tend to surpass AI solutions with regard to the standard of the lexicographic material provided, the quality of the lexical data may not be in itself enough to make users prefer dictionaries edited by lexicographers. A non-trivial issue is that electronic dictionaries that have evolved from print editions are generally stand-alone references that are notably lacking in terms of integration with other systems.

In this session, I will employ practical examples to discuss differences between consulting stand-alone dictionaries and accessing lexical data integrated into other systems from the perspective of behaviour and habit-formation theory. The discussion leads to the inevitable question: who will carry on using dictionaries in the age of AI?



Ana Ostroški Anić

**Institute for the Croatian
Language**



Federica Vezzani

University of Padova

Tools and Methods for Linguistic Research on Neologisms: Evidence from the ENEOLI Survey

Abstract: This talk examines the tools and methods currently used in linguistic research on neologisms, drawing on evidence from the ENEOLI (COST Action CA22126-European Network On Lexical Innovation) survey carried out within Working Group 2, Methods, Digital Resources and Tools for Neology. The survey offers a broad overview of how neologisms and novel senses are identified, analysed, monitored, and documented across different linguistic and professional contexts.

It focuses in particular on the role of corpora, the use of digital tools for extraction and monitoring, and the balance between manual procedures and computational support. It also considers some of the methodological issues that emerge from current practices, including the need to relate linguistic expertise to increasingly diverse digital environments and research workflows. Particular attention is given to the ways in which neologism research develops across lexicographic, terminological, and wider linguistic perspectives.

Finally, the talk reflects on what the ENEOLI survey reveals about the current state of the field and points to the need for more structured resources, clearer methodological frameworks, and stronger coordination in future research on lexical innovation.



Besim Kabashi

University of Tübingen

Non-standard Data in Lexicography: In the Past, and Today, in the Generative

AI Era

Abstract: Non-standard lexical data refers to lexical elements, words, multi-word expressions, and phrases that do not conform to the norms of standard language — usually the formal, generally accepted, and officially recognized forms of a language used in education, the media, and official communication in a country or state.

Non-standard language data deviates from the standard e.g., dialect variants, such as the German “ne” for “nein” (English “no”), slang and informal expressions, youth language, (variants of) Computer-mediated Communication (CMC) / Internet language, Code-mixing and -switching, hybrid forms, contractions, emojis, etc., as well as non-standard spelling or pronunciation.

Importance: Although they are not standard, they are used and make up a significant part of communication. These data can be found in various places and situations-generated and used by a wide variety of users.

Identification, analysis and organization of the data: Identification, analysis, and structured organization are important for understanding speakers and data. This may be the case for scientific purposes, but also for practical everyday use.

Processing and organization of the data: From the perspective of computational lexicography, the processing of this data is no easy task. The difficulties begin with identification, disambiguation, normalization, etc., and continue until the description is complete.

Covered aspects: We will address various aspects of this topic, i.e. from classical views to modern ones in today’s world, i.e., including their machine language processing, or how they are treated in the context of generative AI. In particular, the focus is on the following aspects: Linguistic research and Natural language processing (NLP / AI).



Ilan Kernerman

Lexicala by K

Dictionaries

LexLM: Lexicography-Inspired

Language Models

Abstract: The spirit of legacy lexicography is eternal — born from instinct to explore and understand, need to formulate, and passion to share knowledge, converging the art, and craft, and science of language analysis and representation, prior to dictionaries and forever beyond. This quarter century was dictionaries' golden age — so infinitely abundant and easily accessible — but that might be their swan song: like royal chariots transformed into Volkswagens, when even noble horses become a hobby. Here comes the era of artificial intelligence — artificial, yes, but no intelligence yet, nor heart. As singularity extinguishes human flame, may mighty AI agents be lexinspired.



Robert Lew

Adam Mickiewicz

University

AI Chatbots as an Alternative to Dictionaries: Are They Used, and with What Success?

Abstract: The appearance and growth of LLMs and LLM chatbots has driven many to use them to do things previously done with the help of other tools, and dictionaries are no exception here. In my talk, I examine the recent and current empirical evidence on the extent of the transition for learners of English. Do they still use dictionaries, or do they use chatbots instead; and if so, for what? Next, I report on several studies testing the relative success of dictionaries (monolingual and bilingual) and various chatbots in lexical tasks, both in terms of immediate success and learning (vocabulary retention). These new studies tell us to what extent dictionaries are still useful tools, at least in the language-learning context.



Rute Costa

NOVA University Lisbon

The Role of Knowledge Organisation in Terminological Definition Drafting

Abstract: This paper examines the role of Knowledge Organization (KO) in supporting the drafting of terminological definitions within specialised knowledge domains. In terminology work, definition drafting is not merely a linguistic task but a knowledge-intensive activity that requires structuring, selecting, and representing domain-specific conceptual content in a consistent and operational form. KO systems, including ontologies, taxonomies, classification schemes, and thesauri, provide semi-formalised or formalised structures for organising concept systems, thereby offering a crucial foundation for coherent definition drafting.

By integrating theoretical perspectives from terminology science, specialised lexicography and knowledge organisation, this paper highlights KO as a critical mediating layer between conceptual knowledge and linguistic representation. We conclude that this mediation can improve precision, transparency, and reusability in terminology resources, particularly when definitions must function across both human-centred and machine-readable contexts.

In our presentation, we outline a methodology showing how existing validated ontologies and data extracted from corpora were used in parallel to convert specialised definitions into popularised ones. The resulting definitions are drafted and first validated by domain experts using a dedicated validation protocol. Subsequently, non-experts assess and confirm their comprehension of the popularised definitions, ensuring accessibility and clarity for broader audiences.



Sven Tarp

Aarhus University

AI and Lexicography: The Three Changes, the Five Unchangeables and the One Existential

Abstract: Grounded in recent research on artificial intelligence and the future of humanity (Tarp, 2026), this presentation examines how generative AI is reshaping lexicography at both practical and conceptual levels. It argues that the key challenge is not the technology itself, but how it is understood and framed. In this respect, the frequently asked question “What can AI do for us?” is critically reassessed as incorrect and potentially dangerous.

Drawing on the Function Theory as formulated by Tarp (2008) and further developed by Fuertes-Olivera & Tarp (2014), Nomdedeu-Rull & Tarp (2024), among others, lexicography is approached as an independent discipline focused on producing utility tools that satisfy user needs in specific situations. These needs arise from extra-lexicographic contexts and are shaped by user characteristics such as linguistic competence and subject knowledge, as well as by their experience with the consulted resource. While these principles remain valid, the presentation highlights that AI introduces new opportunities and risks that require a reconsideration of how user needs are identified, how tools are designed, and how lexicographic processes are organized.

At the same time, fundamental requirements remain unchanged. Lexicographers must retain control over their products and rely on a combination of theoretical insight, domain expertise, practical experience, and a deep understanding of human behavior. These competencies cannot be delegated to AI without compromising the discipline.

The presentation adopts a cautiously optimistic perspective on the future of lexicography. However, this optimism is conditional upon the appropriate use of AI. Without a profound understanding of the relationship between human and artificial intelligence, AI may ultimately pose an existential threat to the discipline rather than a sustainable advancement.



陈恋

CHEN Lian

Université d'Orléans

From Corpus to Dictionary: A Computational Lexicography Framework for Neologisms in Hong Kong Written Chinese Based on OntoLex-Lemon

Abstract: This study proposes a computational lexicography framework for the identification, diachronic analysis, and lexical-semantic representation of neologisms in Hong Kong Written Chinese between 2016 and 2026. Conducted within the ENEOLI COST Action (CA22126) and the NeoLex project, the research aims to develop sustainable digital lexical resources for documenting contemporary lexical innovation.

The corpus consists of 732 newspaper articles collected from three major Hong Kong media outlets between 2016 and 2026. After normalization, temporal annotation, and Chinese word segmentation, the corpus was transformed into a diachronically structured dataset suitable for neologism detection and semantic change analysis. A reference lexicon containing 39,988 entries attested before 2016 was used as a lexical baseline. Following the differential approach to neology, lexical units absent from the reference lexicon were considered potential neological candidates.

The methodology is organized into three complementary modules. The first module focuses on formal neologism identification. A character-based sliding-window approach was applied to extract candidate forms directly from the corpus. Linguistic and statistical filtering based on frequency, document frequency, Pointwise Mutual Information (PMI), and left-right entropy was then used to identify lexical units with a high probability of lexicalization.

The second module addresses semantic evolution. Candidate forms were analysed using contextual sentence embeddings generated with multilingual Sentence-BERT. Several indicators were computed, including contextual similarity, semantic dispersion, clustering structure, and yearly semantic drift. The analysis identified more than 2,000 candidates with sufficient diachronic evidence for semantic investigation, representing over 35,000 contextual occurrences. Results reveal different semantic profiles, ranging from stable lexical units to candidates exhibiting significant contextual diversification. Terms associated with major socio-political and technological developments, such as 新冠疫情, 動態清零, ChatGPT, NFT, Web3, and 元宇宙, display particularly strong semantic evolution across the observed period.

The third module focuses on lexical knowledge representation. Building upon the extracted candidates and their semantic profiles, we propose an OntoLex-Lemon-based modelling framework in which validated neologisms are represented as ‘ontolex: Lexical Entry’ resources linked to lexical forms, lexical senses, contextual examples, and corpus-derived metadata. Lexical senses are connected to conceptual entities represented through ‘ontolex: LexicalConcept’, while semantic domains and conceptual relations are organized using SKOS. The model also incorporates diachronic metadata, including first attestation year, frequency measures, PMI, entropy, and semantic drift indicators.

The resulting ontology supports interoperability with Linked Open Data resources and provides the foundation for a dynamic digital dictionary of neologisms in Hong Kong Written Chinese. The study contributes a reproducible pipeline that combines neologism detection, semantic change modelling, and OntoLex-Lemon representation within a computational lexicography framework, enabling the documentation of lexical innovation and semantic evolution in contemporary Hong Kong Written Chinese.



高永伟

GAO Yongwei

复旦大学

Fudan University

How Can AI Facilitate the Compilation of English-Chinese Idiom Dictionaries?

Abstract: This study explores the AI-empowered compilation of English-Chinese idiom dictionaries and examines how recent advances in artificial intelligence (AI) can improve the quality, efficiency, and usability of bilingual lexicographic work. Idioms are among the most culturally loaded and semantically opaque units in language. In English-Chinese lexicography, the treatment of idioms has long faced several recurring problems, including incomplete coverage, vague sense discrimination, insufficient contextualization, overreliance on invented examples, and the lack of up-to-date evidence from authentic media usage. Against this background, the present study investigates how AI chatbots can facilitate the compilation of a more dynamic, corpus-based, and user-oriented English-Chinese idiom dictionary.

The paper first provides a brief historical review of English-Chinese dictionaries of idioms, and then discusses the major challenges in the compilation of bilingual idiom dictionaries. Traditional lexicographic approaches, while valuable, are often labor-intensive and limited by the scale of manual data processing. As language in the media evolves rapidly, dictionary entries may quickly become outdated unless they are regularly revised on the basis of fresh evidence. Although new idioms emerge at a much slower rate than neologisms, the recording of them in idiom dictionaries or general ones is also of great necessity if such dictionaries aim to better meet the needs of their users.

The study then proposes an AI-empowered workflow for idiom dictionary compilation. This workflow integrates large language models, corpus linguistics, and human editorial oversight. First, AI-assisted corpus mining can identify candidate idioms and retrieve authentic examples from newspapers, magazines, online media, literature, subtitles, and other

textual resources. Second, AI can support sense analysis by grouping examples according to meaning, usage pattern, register, and pragmatic force. This helps lexicographers distinguish literal from figurative uses and identify subtle differences among related senses. Third, AI-assisted bilingual alignment can generate preliminary Chinese equivalents, paraphrases, and translation notes, which are then checked and refined by human editors.

A central argument of this study is that authenticity and contextual richness should be core principles in idiom dictionary design. AI makes it possible to move beyond decontextualized definitions by providing abundant real-world examples, especially from contemporary English-language media. For Chinese users, an effective dictionary entry should ideally include the head idiom, Chinese equivalents, notes on register and style, variant forms, common collocations, and one or more authentic examples with accurate Chinese translation. AI can help generate draft components for all these fields, while human experts ensure precision, cultural appropriateness, and pedagogical value.

The paper also discusses risks and limitations. AI systems may hallucinate, misidentify idioms, confuse literal and figurative uses, or produce unnatural Chinese translations if left unchecked. Therefore, the study emphasizes a human-in-the-loop model in which lexicographers evaluate sources, verify examples, and make final editorial decisions.

In conclusion, AI-empowered compilation offers a promising new path for English-Chinese idiom dictionaries. It can expand coverage, improve timeliness, enrich usage notes, and enhance the descriptive accuracy of entries, while reducing repetitive manual workload. However, the success of this approach depends on a balanced collaboration between computational tools and professional lexicographic judgment, or the right mix of AI and HI (human intelligence).



耿云冬

GENG Yundong

华中农业大学

Huazhong Agricultural

University

Toward Human-AI Collaboration in Building a Chinese-English Lexicographical Database for Agricultural Heritage

Abstract: Agricultural heritage terminology is culturally embedded, domain-specific, and often under-resourced in bilingual lexicographical practice, which poses challenges for concept modelling, equivalent selection, definition writing, and user-oriented access. To address these challenges, this paper develops a human-AI collaborative framework for constructing a Chinese-English lexicographical database of agricultural heritage terminology. The proposed resource is conceptualised not as a conventional bilingual dictionary, but as a database-oriented lexical knowledge system linking terms, concepts, equivalents, definitions, multimodal evidence, and multiple access routes. The framework consists of three interdependent components: knowledge graphs for representing concepts and semantic relations; an instruction-tuned large language model for evidence-constrained drafting of definitions and usage notes under human editorial supervision; and hybrid retrieval for cross-lingual search and exploratory access. Its main contribution is design-oriented, as it translates core lexicographical functions into a computational workflow and specifies points of human validation throughout data collection, modelling, drafting, revision, and retrieval. The study demonstrates that, in culturally embedded and under-resourced domains, AI is most effectively deployed not as an autonomous “lexicographer”, but as a constrained assistant embedded within an evidence-based, database-oriented lexicographical architecture.



胡文飞

HU Wenfei

四川外国语大学

Sichuan International

Studies University

汉英词典儒学词研究：范式革新与编纂实践

摘要：词典是社群语言的结晶，承载着民族的文化记忆。本研究以文化记忆理论为框架，融合语料库驱动、语料库翻译学及对比短语学等多元路径，推进词典学研究范式的创新。在编纂实践层面，本文运用常态与拓展模式、短语驱动原则、系统适应原则及文化拓扑原则，系统优化汉英词典中儒学词的表征模式。基于范式探索，研究团队完成了《汉英儒学成语词典》的编纂工作，并基于大语言模型技术启动《汉英儒学词典》的编撰项目，旨在解决汉英词典编纂中的语义描写单一化、文化阐释碎片化及编纂技术更新滞后等关键问题。



亢世勇

KANG Shiyong

鲁东大学

Ludong University

现代汉语汉语融媒（体）学习词典构建 实践探索

摘要：融媒（体）词典是近年来辞书领域关注的热点。就语言学习来看，编纂融媒（体）汉语学习词典是非常必要的。2018年以来，我们在国家社科基金项目、国家语委科研项目的支持下，我们编纂了面向基础教育的融媒体词典《当代汉语学习词典》和《现代汉语新词语融媒体词典》。融媒（体）词典的基本特征是三融合，即多媒体、多模态信息的融合，辞书相关资源的融合，辞书编纂者与用户之间互通融合。学习词典的基本原则是“坚持以用户为中心的原则”，即适应学习者的认知水平和能力，满足用户的学习需求。为此，在编纂《当代汉语学习词典》时，为了词典释义等通俗易懂，让学习者能够看得懂，文字、图片、视频、音频等多媒体、多模态信息融合，协同呈现相关信息；为了满足学习者的需要，打破字典与词典的界限，打破各类词典（同义词词典、反义词词典、分类词典、词源词典、图解词典等）的界限，从字的形音义之间的关系，到语义构词规律、词义的引申发展、词义的文化背景、同义词辨析、关联语汇等全面呈现，希望该词典能够解决学生学习的大部分问题。在编纂《现代汉语新词语融媒体词典》时，搜集广泛的例句说明每个词语的语义、语法、语用特点，每个例句后附有该例句出处的网址，通过网址可以了解到词语报纸、电视、网站等平台的信息，实现多媒体、多模态信息融合的特征。同时，还附有每个词语的词源及社会文化背景，讲清楚该词语的产生发展历程。两部词典在词频统计、释义、例句选择、文化背景阐释等方面都充分利用了大模型，不仅速度快、效率高，而且准确性、适配性都有较好的保证。



柳长青

LIU Changqing

广东外语外贸大学

Guangdong University of

Foreign Studies

Research on the Intelligent Compilation System of Generative AI-driven Dictionary and Human-computer Collaboration Mechanism: Take Loanwords in Chinese as an Example

Abstract: The article focuses on the application of generative AI in dictionary compilation, and puts forward an intelligent compilation system and human-computer collaboration mechanism for loanwords in Chinese. Unlike the traditional model of corpus-driven or single text generation, this article reconstructs the dictionary compilation process into a controllable process composed of template constraints, semantic generation and quality evaluation. The system takes the large language model as the core engine, realizes the structured generation of etymology, interpretation and example sentences through the template prompt mechanism, and combines automatic scoring and abnormal detection modules to carry out computable quality control of the generated results. On this basis, the human-computer collaborative closed loop of “generation-evaluation-correction-regeneration” is introduced to achieve continuous optimization of entry quality. At the same time, the system supports multi-modal information expansion, so that dictionary entries have the ability to evolve from text interpretation to multi-dimensional semantic expression. Experiments taking loanwords in Chinese as a case show that this method is superior to the traditional compilation process in terms of generation efficiency, structural consistency and semantic stability. Research shows that generative AI can promote the transformation of dictionaries from static text resources to dynamic, computable knowledge systems.



杨玉玲

YANG Yuling

北京语言大学

**Beijing Language and
Culture University**

An Empirical Study on the Large-Scale Generation of Multimodal Resources Driven by Structured Data

Abstract: Confusing word discrimination has long been a key and challenging issue in second-language vocabulary learning and the compilation of export-oriented learner's dictionaries. Previous studies have confirmed that learners have a strong demand for confusing word discrimination resources in learner's dictionaries (Zhang, 2011; Yang & Yang, 2019). Traditional dictionaries provide only a limited number of discriminations and rely predominantly on text-based explanations, which impose heavy cognitive load and lack readability and attractiveness. By contrast, multimodal presentation can effectively reduce learners' cognitive burden and improve the efficiency and overall experience of vocabulary acquisition (Zhang, 2021). Nevertheless, the manual development of multimodal discrimination content involves high costs and long production cycles, making it difficult to meet the practical requirements for large-scale and batch development of dictionary resources. Generative artificial intelligence offers a novel technical approach to the automated and scalable production of smart dictionary content (Cooper, 2023; Huang et al., 2023).

Taking 276 groups of Chinese confusing words as research samples and adopting the perspective of integrated media dictionary compilation, this study empirically explores the enabling effect of structured data on the large-scale generation of multimodal interactive discrimination content from three

dimensions: generation efficiency, content quality, and human-computer interaction experience. A structured-data-enabled experimental group and an unstructured-data control group were set up for comparative analysis. The comparative ratios of the two groups are as follows: valid generated items (177:99), average revision rounds (2.09:4.85), prompt length (63.49:159.2 words), average time consumption (7.59:14.68 minutes), discrimination quality score (4.47:3.63), and emotional experience value (-0.1:-0.6). The empirical results reveal that structured data significantly improves the generation efficiency and content quality of multimodal confusing word discrimination, optimizes human-computer collaborative experience, and maintains stable performance in large-scale batch generation. This approach effectively resolves the prevalent practical dilemma in integrated media dictionary compilation that “single entries perform well while batch generation suffers significant degradation”. This study provides empirical evidence and a feasible technical paradigm for data-driven content production of integrated media dictionaries and the construction of AI-assisted lexical discrimination resources.



张永伟

ZHANG Yongwei

中国社会科学院语言研究所

Institute of Linguistics,

CASS

Research on Evaluating the Intelligent Compilation Capabilities of Large Language Models for Chinese Lexicographical Dictionaries

Zhang Yongwei, Ma Qiongying, Tong Lijun

Abstract: To thoroughly explore the potential of large language models (LLMs) in Chinese lexicographical compilation, this study constructs an evaluation framework encompassing five core tasks of dictionary compilation: lexical entry selection, part-of-speech tagging, sense division, definition selection and example selection. Referencing the compilation and revision practices of the *Contemporary Chinese Dictionary* (CCD), we developed the Chinese Lexicographical Compilation Evaluation Corpus (CLCEC) and performed automated assessments on ten mainstream LLMs based on this corpus. The experimental results show that LLMs deliver excellent performance in grammatical and semantic comprehension tasks including part-of-speech tagging and sense division. Nevertheless, their capabilities still need further improvement for tasks such as lexical entry selection, which are highly dependent on professional linguistic knowledge and stringent lexicographical specifications. Different LLMs present significant disparities in lexicographical compilation performance, among which Gemini and Claude achieve superior overall results. This study systematically analyzes the specific performance and capability boundaries of various LLMs in lexicographical tasks, offering new references for LLM-empowered Chinese dictionary compilation.

平行论坛发言摘要
Abstracts of Parallel
Sessions

A Human-Centered Perspective to the Integration of AI into Lexicography and Terminology Work

Antonio San Martín

Abstract: Artificial intelligence has the potential of reshaping lexicography and terminology work. The advanced linguistic capabilities of large language models create both opportunities and concerns. While AI can support a wide range of lexicographical and terminological tasks, it also fuels fears that the expertise of lexicographers and terminologists may be marginalized and that the resources they produce could be perceived as redundant.

These two professions thus stand at a critical juncture. What is at stake is not only their role, but also the essential function of dictionaries, termbases, and other language resources in supporting human communication. Mainstream AI development has largely prioritized productivity and cost reduction, often sidelining human well-being. In response, this presentation argues that the integration of AI into lexicography and terminology work should be grounded in the paradigm of human-centered artificial intelligence (HCAI), which places human needs and values at the core of technological deployment.

A fundamental principle of HCAI is that high levels of automation are compatible with strong human control. AI should be conceived as a tool that augments human capabilities rather than replaces them. Human and artificial intelligence have different yet complementary strengths, and the best outcomes arise when they are effectively combined. It follows that lexicographers and terminologists can fully benefit from automation, provided they retain control over the entire workflow, including the choice of tools and the degree and timing of automation.

From a human-centered perspective, AI must contribute to the well-being of all those affected by its deployment. In this context, that includes lexicographers and terminologists, as well as society at large, given the essential role of language resources in human communication. However, in productivity-driven organizational environments, excessive

automation may confine these professionals to the rapid validation of AI-generated output, a trend already documented in translation. Such a shift would reduce professional autonomy and job satisfaction and potentially degrade the accuracy of language resources.

Lexicographers and terminologists must retain a central role in safeguarding the quality and reliability of language resources. The integration of AI into lexicographic and terminological workflows introduces risks of bias that can only be effectively mitigated through professional expertise. These include linguistic bias, such as the influence of English on outputs in other languages; cultural bias, particularly the privileging of Anglo-American perspectives; and social bias, including stereotypical or discriminatory representations. If such biases become embedded in language resources, they may subsequently permeate human communication, thereby eroding linguistic and cultural diversity and adversely affecting certain social groups.

This presentation outlines a research agenda for AI-assisted lexicography and terminology grounded in human-centered principles. Ultimately, the question is not whether AI will be used in lexicography and terminology work, but under which conditions and according to which guiding principles. A human-centered approach positions lexicographers and terminologists as key agents who retain control over AI-assisted workflows, fully harness the advantages of automation, and use them to preserve the socioeconomic functions of dictionaries, termbases, and other language resources.

Analyzing Botanical Definitions in *Maximino Martínez's Catalogue of Mexican Plants*: A Hybrid Entropy-Based Model for Genus-Differentia Decomposition and a Lexicographical Ontology to Build Validated Knowledge Graphs

Carlos F. Diez Sánchez
Alfonso Medina Urrea

Abstract: This paper presents two complementary computational methodologies to analyze the lexicographical definitions in *Maximino Martínez's Catálogo de plantas mexicanas* (CPM, 1923), a foundational dictionary in Mexican botany that documents the vernacular names and descriptive properties of the country's diverse flora.

The first methodology introduces a hybrid model to automatically decompose intentional definitions into their two Aristotelian components: the proximate genus (genus proximum, an “is-a” relationship) and the specific difference (differentia specifica, a “has-a” relationship). The core function relies on two highly accurate rule-based heuristics and an entropy-based algorithm that uses corpus-wide word frequencies to calculate Shannon's entropy within increasing context windows. This identifies shifts in information content to mark the boundary between components. The algorithm is enhanced by a Multi-Class Logistic Regression model, trained on a curated vocabulary of Spanish adjectives as features to dynamically select the optimal entropy threshold. This hybrid approach demonstrated high overall performance, achieving an F1-score of 0.9807 for exact boundary matching on a gold-standard dataset of 7,744 manually labeled definitions, offering a robust tool for lexicographical structure extraction at scale.

The second methodology centers on a comprehensive, data-informed lexicographical ontology used as a structural schema for Knowledge Graph (KG) generation. The ontology comprises nodes for general characteristics (i.e., type of plant, habitat, uses), and specific organs (i.e., leaves, flowers, fruits), which served as a baseline to produce a statistical report of the most frequent botanical terms. The schema was subsequently fed into a Small Language Model (SLM, Gemma 3, 4B) to populate KGs for all 12,860 intentional

definitions. The SLM's performance was evaluated through a dual validation process: first, by benchmarking its output against the statistical baseline, and second, by accepting rare botanical terms identified by the SLM that satisfied a strict substring match constraint. This ensured all extracted values were non-hallucinatory and allowed for the discovery of valid nuances beyond high-frequency terms.

Results show that out of 12,855 entries processed, 8,828 KGs (68.6%) were fully valid, while 4,025 (31.3%) were partially valid (meaning at least one instance failed the substring match constraint). Despite these challenges, the validated data yielded high extraction rates: 91.8% for the 'type of plant' attribute, and 91.6% for the rest of general attributes. The data also confirms the dominance of *planta* ('plant', in 28.4% of the definitions), *árbol* ('tree', 17.8%), and *arbusto* ('bush', 13.9%) as primary classifications, and *hoja* ('leaf', 35.1%), *flor* ('flower', 24.9%), and *fruto* ('fruit', 22%) as essential descriptive features.

To ensure taxonomic consistency with contemporary standards, the CPM was updated by mapping the scientific names to the World Flora Online (WFO) database. This integrated framework — combining information theory for structural parsing and SLMs for grounded knowledge extraction — reflects the informational architecture of the dictionary. The resulting ontology, statistical report, validated KGs and genus-differentia decomposition will be made available on the website: <https://catalogodeplantasmexicanas.colmex.mx/>.

De l'HTR au XML-TEI: un Workflow Collaboratif pour L'édition Numérique de Dictionnaires Chinois-Européens

陈凤仪 CHEN Fengyi
Vincent Paillusson

Abstract: Le projet ChEDiL (ANR-23-CE27-0008)¹ a pour objectif de recenser et numériser des dictionnaires sino-européens produits entre la fin du XVI^e siècle et le début du XIX^e siècle, le plus souvent par des missionnaires occidentaux. Les objectifs numériques du projet sont:

- L'application de méthodes de Handwritten Text Recognition (HTR) à des corpus historiques multilingues,
- L'exploration numérique de ces documents lexicographiques.

Cet abstract présente un workflow pour l'édition numérique collaborative à l'aide de méthodes semi-automatiques. Il s'inscrit dans une démarche reproductible en trois étapes principales:

1. la segmentation-transcription: HTR,
2. la pré-éditorialisation: le format Open Document Text (ODT),
3. l'édition: XML-TEI.

L'étape de HTR s'appuie sur la plateforme Escriptorium² qui fournit une segmentation en lignes et en régions, ainsi qu'une transcription du contenu textuel, exportées dans le format standard ALTO XML.

Dans le cadre d'un projet d'édition numérique académique, l'objectif est d'obtenir un document structuré en XML-TEI. Pour les dictionnaires traités, nous utiliserons la

¹ Dictionnaires sino-européens: manuscrits lexicographiques pour l'étude historique des échanges entre la Chine et l'Europe (fin du XVI^e s.-début du XIX^e s. Agence nationale de la recherche. Disponible sur: <https://anr.fr/ProjetANR-23-CE27-0008> (Consulté le: 27 février 2026).

² Alix Chagué, « eScriptorium : une application libre pour la transcription automatique des manuscrits », Arabesques [En ligne], 107 | 2022, mis en ligne le 13 octobre 2022, consulté le 28 avril 2026. URL: <https://publicationsprairial.fr/arabesques/index.php?id=3100>

spécification TEI-Lex03³. S'il existe des démarches⁴ pour convertir l'ALTO en TEI, cette automatisation ne permet que de récupérer ce qui est attendu par le script XSLT. Par ailleurs, le passage en TEI soulève la question de l'éditorialisation et de l'enrichissement des contenus textuels par un public non spécialiste du XML.

Pour répondre à cette problématique, nous nous appuyons sur une démarche, éprouvée dans plusieurs projets^{5, 6}, consistant à utiliser le format Open Document Text, associé à LibreOffice, pour éditorialiser les dictionnaires.

Le format ODT se présente sous la forme d'une archive ZIP manipulable, elle est composée de fichiers XML dont le schéma est connu. L'application de styles aux paragraphes et aux lignes nous permet de structurer le contenu sans expertise XML préalable.

Afin de faciliter au maximum le travail des éditeurs, des prétraitements automatiques, consistant à styler automatiquement le document produit, seront effectués entre nos fichiers ALTO et leur conversion en ODT :

- le stylage de paragraphe correspond à la macrostructure segmentée à l'aide d'un modèle Yolo⁷ entraîné spécifiquement;
- le stylage de caractère correspond à la microstructure et permettra de distinguer les différentes langues, ainsi que la prononciation, au sein de chaque entrée de dictionnaire. Il repose en partie sur une reconnaissance des entités nommées (NER) entraînée spécifiquement avec spaCy⁸;
- l'insertion de commentaires regroupe l'ensemble des annotations critiques introduites à l'aide d'un vocabulaire contrôlé.

³ Laurent Romary, Ana Salgado, Boris Lehečka, Axel Herold , Piotr Bański, Charly Moerth, Fahad Kahn , Kinga Sramó, Daniel Schopper, Veronika Engler, Jack Bowers, Alexander Geyken, Tomaž Erjavec, Andreas Witt, Snežana Petrović, Jesse de Does, Katrien Depuydt, Mohamed Khemakhem, Vera Hildenbrandt, and Petra Bago. TEI Lex-0 — A baseline encoding for lexicographic data — Clarity for lexicographers. Stability for data. Compatibility for the future. Disponible sur: <https://lex-0.org/> (Consulté le: 27 février 2026).

⁴ Chiffolleau, F., Chagué, A. et Scheithauer, H. (2022) « alto2tei, an XSL Transformation to transform XML ALTO into TEI XML ». Disponible sur: <https://github.com/TEI4HTR/alto2tei> (Consulté le: 27 février 2026).

⁵ Metopes-Outils (sans date). Disponible sur: <https://www.metopes.fr/metopes-outils.html> (Consulté le: 27 février 2026).

⁶ Glorieux, F. (2015) « Le traitement de textes (odt) pour produire des documents structurés (XML/TEI) — Odette », J'attends des résultats, 8 avril. Disponible sur: <https://doi.org/10.58079/tox>.

⁷ Ultralytics, YOLOv8, <https://huggingface.co/Ultralytics/YOLOv8>

⁸ Matt Honnibal, Ines Montani, spaCy, <https://github.com/explosion/spaCy>

Le texte en ODT peut alors être édité puis être converti en XML-TEI en vue de son exploitation finale (édition papier, publication web, conversion en Ontolex-Lemon, etc.).

Ainsi, ce workflow constitue une expérimentation méthodologique qui favorise l'accessibilité des outils d'annotation ainsi que le travail collaboratif d'équipes pluridisciplinaires non formées au XML pour la production de données éditoriales lexicographiques.

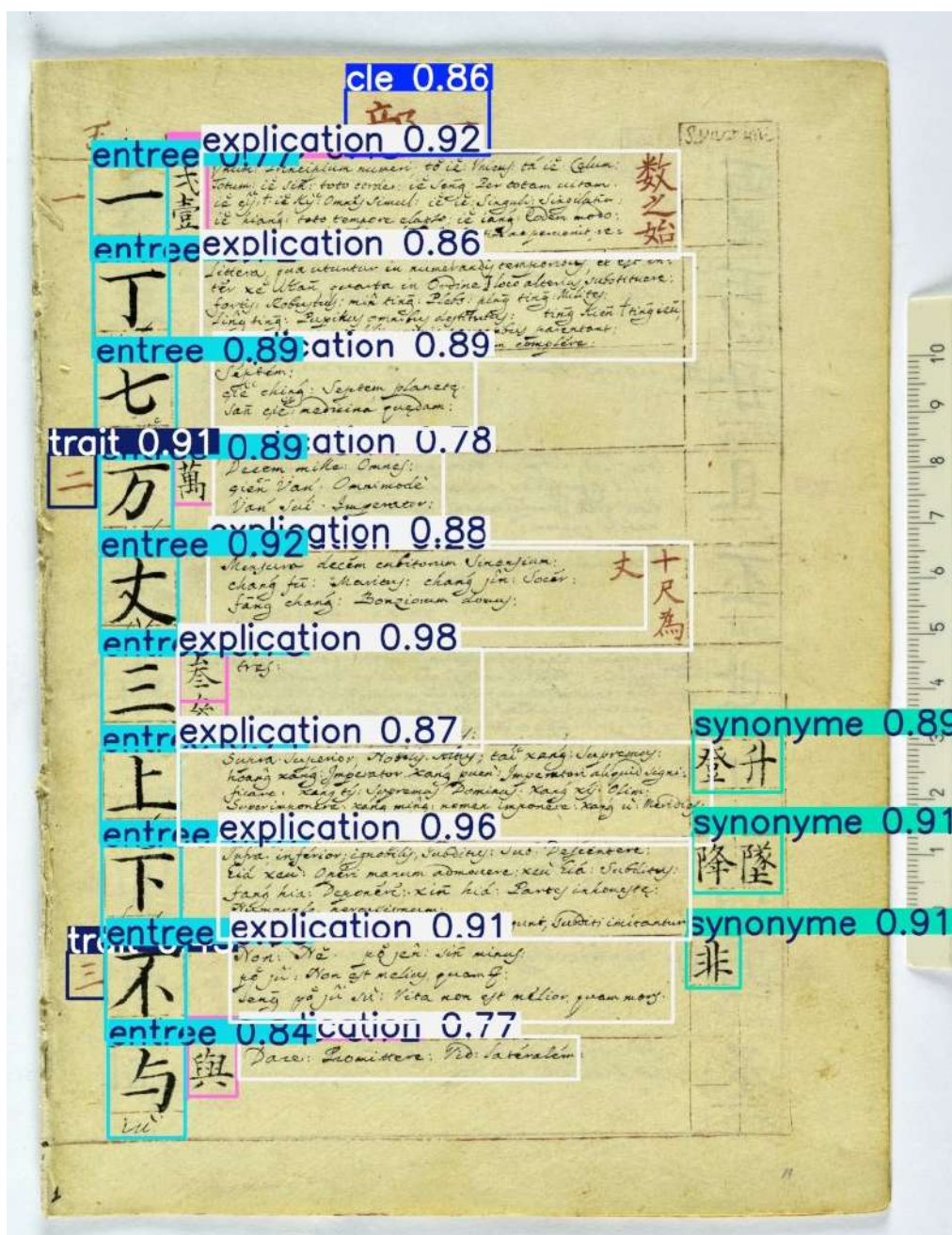


Figure 1: Visualisation d'une page segmentée avec Yolov8 selon la macrostructure - BNCF Rinuccini 22

Times New Roman 12 pt G I S S x² x₂ A A

Commentaires

Styles

Afficher les aperçus Mise en

Styles personnalisés

Abbréviation

Annotation_auteur

Annotation_chercheur

Caractère chinois

Caractères variantes

Exemples_chinois

Explication_latin

Insertion

Mot-biffé

Pinyin

Synonyme

Transcription phonétique

Tag : nombre

Nom du chercheur

27/02/2026

一 一 式 壹 乙

Unum: Principium numeri: *tō iē*: Unicus: *tō iē*: Celum: Totum: *iē sū*: toto corde: *iē sēg*: Per totam vitam: *iē cī* vel *iē kī*: Omnes simul: *iē iē*: Singuli: singulati: *iē hiang*: toto tempore elapso: *iē tang*: Eodem modo: Statim ac: *iē tō*, *cien tōt*: Statimac pervenit, re: vērsum ēst: 數之始 *shū chī shū*

丁 丁 丁

Littera, qua utuntur in numerandis temporibus, ēst in: *tēr xē Kan*, quarta in ordine] loco alterius substituere: fortis: Robustus: *min tīng*: Plebs: *ning tīng*: Milites: *hīng tīng*: Pupillus omnibus destitutus: *tīng Kien* vel *tīng iēn*, tempus triennii, quo filii mortuis parentibus parentant: *ching tīng*: 16^m vel 18^m etatis annum complere

七 七

Septem: *qī chīng*: Septem planeta: *san qī*: medicina quaedam:

Figure 2: Exemple d'une page ODT stylisée - BNCF Rinuccini 22

Comparing the Efficacy of an LLM-based Chatbot, Machine Translation App, and Bilingualized Learner's Dictionary in Receptive and Productive Lexical Tasks: Success Rates, Lookup Speed, and Vocabulary Retention

冯永芳 FENG Yongfang

Abstract: Digital lexical tools such as ChatGPT, Youdao (a machine translation app), and *Oxford Advanced Learner's Dictionary (OALD, 10th bilingualized edition)* can support L2 vocabulary learning. However, a comprehensive comparison of their efficacy across receptive and productive vocabulary tasks remains underexplored. This study investigated their relative effects on learners' vocabulary learning in both task types, assessing success rates, lookup speed, and vocabulary retention, alongside learners' perceptions. A total of 213 Chinese university students at the CET-4 proficiency level were randomly assigned to three groups of 71 participants each. They completed receptive (noun meaning recall) and productive (phrasal verb form recall) tasks using their assigned tool, followed by an immediate posttest, a delayed posttest one week later, and a perception survey. Results revealed that Youdao had the lowest success rates but was the fastest tool in the receptive task, though the slowest in the productive task. In contrast, ChatGPT demonstrated superior efficiency in the productive task. In the immediate posttest, OALD10 and Youdao users outperformed ChatGPT users in meaning recall, with OALD10 users also surpassing Youdao users in form recall. However, all groups exhibited poor delayed retention across tasks. Perception data further highlighted OALD10's authority and support for deep learning, despite occasional information overload; Youdao's multifunctionality, though criticized for excessive advertisements; and ChatGPT's efficiency and adaptability, though concerns over accuracy and hallucinations persisted. These findings provide implications for English pedagogy and the design of digital lexical tools.

Evaluating LLMs in the Lexicographic Detection and Treatment of Chinese Neologisms: Evidence from Large Chinese-English Dictionaries

黄锦鸿 HUANG Jinhong

Abstract: Contemporary large Chinese-English dictionaries (LCEDs) in mainland China, such as *New Age Chinese-English Dictionary* (2014, NACED), *New Century Chinese-English Dictionary* (2025, NCCED), *The Century Chinese-English Dictionary* (2021, TCCED), and *The Chinese-English Dictionary* (Unabridged, 2015/2024, CEDU) increasingly attach importance to the detection and description of Chinese neologisms (Zhao 2016; Gao 2021). Nevertheless, effective monitoring of Chinese neologisms remains challenging due to the lack of balanced, large-scale, and dynamically updated Chinese corpora, which weakens the empirical basis for Chinese-English (C-E) dictionary compilation and revision. Reliance on manual neologism collection is labour-intensive and time-consuming, thereby constraining the efficiency and timeliness of lexicographic updates.

The present-day LCEDs vary substantially in their inclusion policies, translation practices, and example selection for new lexical items. First, such variation is evident in the coverage of 爆 ('to explode; to go viral') related neologisms (e.g. '爆款 bestseller' '爆仓 be overloaded' '爆红 become popular overnight'), many of which are unevenly included among the said dictionaries and susceptible to omission. Second, neologism translation remains problematic: inaccurate (e.g. '打工人 drudge; labourer; worker bee' in NCCED) or verbose English equivalents (e.g. '海选 wide selection from a large number of candidates' in TCCED), omission of newly emerging senses (e.g. '水课 gut course' in CEDU), and insufficient or excessively detailed explanatory notes (e.g. '爆炒 A metaphor for launching a whirlwind mass media advertising campaign for quick and optimum commercial impact in a short period' in TCCED). Third, examples are missing in some of the neologisms (e.g. '洗白 whitewash' and '垃圾评论 spam comment' in NCCED). These issues raise the question of how neologisms can be better treated in LCEDs.

Approaches to automatic entry compilation have mainly relied on rule-based methods (Wei and Yuan 2014) and deep learning based methods (Noraset et al. 2017; Gadetsky et al.

2018; Kong et al. 2022). Nowadays, generative large language models (LLMs) have opened up new possibilities for dictionary compilation. Studies on dictionary automation and implications of LLMs for lexicography have gained momentum (Rundell 2023; Fuertes-Olivera 2024; Rees and Lew 2024; de Schryver 2025; Han & de Schryver 2026), while a smaller body of work has explored their use in neologism detection and treatment (e.g. Nam et al. 2025; Wu et al. 2026). The application of LLMs to Chinese neologisms in LCEDs is still underexplored. The present study therefore investigates this issue by comparing the neological entries generated by four LLMs (i.e. ChatGPT, Grok, DeepSeek, and ERNIE Bot) with the corresponding entries in CEDU, NACED, NCCED as well as TCCED, employing a hybrid evaluation approach that integrates human assessment and automated evaluation.

The present research addresses the following questions: (1) To what extent do LLMs effectively detect Chinese neologisms across heterogeneous data sources beyond traditional corpus-based approaches? (2) How well do LLMs perform in terms of generating appropriate English equivalents, differentiating senses, and selecting illustrative examples for Chinese neologisms in LCEDs? (3) What comparative strengths and limitations do different LLMs exhibit when evaluated against neological entries in existing LCEDs?

Methodologically, the study operationalises LLM-assisted neologism detection through a multi-step workflow. First, neologism candidates are collected from heterogeneous sources including online news reports, social media platforms, web forums, and academic texts. Second, prompts are designed to require LLMs to identify candidate neologisms, generate English equivalents, distinguish emerging senses, and provide illustrative examples. To reduce hallucinated outputs, all LLM-generated entries are cross-checked against attested Chinese usage data, existing LCED entries, and contextual evidence from authentic texts. Entries lacking sufficient usage evidence or contextual support are excluded from the analysis.

A multi-dimensional comparative framework is proposed to assess LLM performance at macrostructural and microstructural levels. From a macrostructural perspective, the study assesses the ability of LLMs to identify Chinese neologism candidates from diverse sources (news, academic journals, social media, and web-based texts etc.), thereby addressing the

long-recognised news-centred bias in Chinese neologism detection. Microstructurally, the analysis focuses on the capacity of LLMs to provide appropriate English equivalents, distinguish emerging senses, and generate illustrative examples.

Rather than advocating the replacement of human lexicographers, this study situates LLM-assisted methods within established C-E lexicographic traditions and argues for a reconfigured workflow in which LLMs function as exploratory and supportive tools, especially in the early stages of neologism detection and draft entry generation. This study contributes to AI-powered lexicography by demonstrating the potentiality and applicability of LLMs to enhance the coverage, accuracy, and efficiency of Chinese neologism treatment in LCEDs. It highlights the need for systematic LLM evaluation and high-quality lexical data resources, while reaffirming the essential role of human editorial judgment in bilingual dictionary compilation.

Sense Induction and Sense Frequency Statistics of Chinese Polysemous Words Based on Large Language Model

李芳棋 LI Fangqi

Abstract: Word sense induction and sense frequency statistics are fundamental to dictionary compilation yet remain challenging due to subjectivity and the inefficiency of manual corpus analysis. This study proposes a corpus-driven, human-in-the-loop workflow based on a large language model (ERNIE X1) accessed via API calls, and evaluates its effectiveness on eight Chinese polysemous words of varying semantic complexity (经过 jīngguò, 猴 hóu, 牢 láo, 可以 kěyǐ, 坐 zuò, 火 huǒ, 把 bǎ, 打 dǎ). The workflow comprises five phases: corpus preprocessing, autonomous sense induction via semantic clustering, three rounds of LLM self-optimization, manual sense adjustment (splitting/merging) with lexicographic criteria, and LLM-based sense frequency annotation validated against manual annotation. Results show that for words with well-differentiated senses (经过 jīngguò, 猴 hóu, 牢 láo, 可以 kěyǐ, 火 huǒ), the LLM's autonomous induction fully covered all dictionary senses. For highly complex words (把 bǎ, 打 dǎ), three optimization rounds achieved 85.71% and 77.14% satisfaction, and manual intervention brought both to 100% satisfaction. The LLM also discovered plausible new senses. In frequency annotation, most words reached “almost perfect” agreement ($Kappa \geq 0.85$) with human annotators. Limitations include output variability, API instability, overgeneralization, and difficulty controlling sense granularity. The study demonstrates that LLM-assisted workflows can substantially improve the efficiency and objectivity of sense induction and frequency statistics, providing reliable support for dictionary revision, new sense discovery, and sense ordering. Future work should focus on lexicography-specific LLMs and standardized sense description frameworks.

Lost to History? Archiving Open-Data Dictionaries for Future Enquiries and Research Reproducibility

Louis Lecailliez

Introduction

Open-data dictionaries and corpora are the basis of plenty NLP tools and applications, as well as research papers. Currently available resources may be evolving after release – sometimes at a very slow pace – instead of being static artifacts. Not only the content itself is subject to additions and corrections, the structure of data or file format can also change during the lifetime of a project.

We can cite for instance the schema change of the XML-encoded Chinese-French open data dictionary CFDICT⁹ that happened between 2011 and 2016. Moreover, it also underwent vast changes in its number of entries, ranging between 207,966 (in 2011), 6,209 (2016), 44,497 (late 2017) and 56,296 (2024). We noted at least 3 other cases, including the Chinese monolingual dictionary *Revised Mandarin Chinese Dictionary* (重編國語辭典修訂本 *Zhòng biān guóyǔ cídiǎn xiūdìng běn*) and the web version of Hatoma dialect dictionary (Kajiku, 2020), during our project spanning over 7 years that use an important number of open dictionaries.

Problems

Changes between versions often aren't documented, versioned nor made explicit. This make adjustment for downstream applications necessary as newer versions of data can't be used as drop-in replacement of the previous ones. In crowd-sourced dictionaries (Čibej, Fišer & Kosem, 2015), rapid evolution and constant change is a core feature of a project. Some platforms such as Jibiki (Mangeot, 2018) keep tracks internally of contributors' changes but don't expose them.

In addition, resources may simply disappear or become inaccessible because of link rot (Goh & Ng, 2007). As of writing, the aforementioned CFDICT file is inaccessible from its official website. This happens despite licensing allowing for reproduction and distribution: if those rights aren't put in effect by users, data may be lost, possibly forever.

⁹ <https://chine.in/mandarin/dictionnaire/CFDICT/>

Scientific reproducibility is also impacted: literature mentions resources by name but content of said resources may well have been modified or could have disappeared since publication. In effect, there is no history for most open data dictionaries. The same issue of lack of history, impacting future research on dictionary evolution in the digital age, was brought to light by Rees (2025) for commercial web-based commercial monolingual English dictionaries. While some of the issues raised equally applies to open-data dictionaries, their open nature allows us to directly try to address the issue ourselves, without being withheld by copyright issues.

Proposed Solution

We propose a semi-automated solution structured as a 3-tiers project to collect and archive over time East-Asian open dictionaries. The first tier is a manifest repository containing text files informing both human and machines of existing lexicographic resources, where to find them and their expected frequency change (daily, monthly, etc.)

The declaration files are used by the second tier of the system: software implementations that attempt downloading resource files at an interval specified. The third tier is the storage space (a GIT LFS repository) where the retrieved files are physically stored. When the downloading agent can't find a file, it sends a notification to maintainers so that potential link rot is detected early.

This solution gives awareness of when a resource disappears and provide historical data that can be used for future research on their evolution. Some past files we acquired during years of research will also be added to the repository storage.

Low-Resource Sepedi Corpus Engineering for Computational Lexicography: A Data-to-Dictionary Pipeline with Compact LLM-Assisted Example Generation

Mathews Tladi

Abstract: Computational lexicology and lexicography increasingly rely on large, representative, and reusable text corpora developed through transparent, reproducible preprocessing and evaluation methods. For many African languages, including Sepedi, also known as Sesotho sa Leboa, digital text remains dispersed across heterogeneous sources, with inconsistent encoding, duplication, and frequent code-mixing. These issues reduce its suitability for corpus-driven dictionary compilation and for training modern NLP systems. This paper presents a reproducible Sepedi data-to-dictionary pipeline that transforms publicly accessible text into dictionary-facing resources, including frequency-ranked headword candidates, n-gram and collocation lists, multiword expression evidence, and candidate example sentences.

The proposed pipeline includes source discovery and consolidation, encoding normalisation, boilerplate and markup removal, deduplication, and noise reduction involving punctuation artefacts, emojis, Unicode symbols, hashtags, and excessive whitespace, while preserving Sepedi orthography and diacritics such as “š”. Stop words are retained because they preserve grammatical fluency and support authentic example sentence selection. To reduce English–Sepedi code-mixing, a lexicon-informed filter excludes sentences that are predominantly English, using an 80% English-lexicon coverage threshold with at least two tokens. This threshold is informed by manual spot checking to reduce false positives involving named entities and common borrowings. The pipeline is batch-based and reproducible from raw source texts to a single cleaned dataset. Data governance is also addressed by restricting the corpus to publicly accessible and research-approved sources, with licences verified, complied with, and properly attributed.

The resulting corpus contains 6,429,274 tokens and 138,470 unique word types, providing broad lexical coverage for the Sepedi language. Dictionary-facing outputs are derived from the corpus to support both macrostructural and microstructural lexicographic

decisions. These outputs include frequency-ranked word lists for headword prioritisation and lexical coverage auditing, n-gram and collocation candidates for phraseology and multiword-expression discovery, and filtered pools of candidate example sentences for entry microstructure, particularly usage examples, register observation, and contextual meaning. The frequency patterns show a steep, Zipf-like distribution dominated by high-frequency function words, offering useful evidence for lexical prioritisation, Sepedi language coverage auditing, and dictionary planning.

As a secondary proof of use, a compact transformer language model, DistilGPT-2, is fine-tuned using Low-Rank Adaptation (LoRA) to support LLM-assisted example-sentence drafting, with fixed training, validation, and test splits of 80:10:10. The model is evaluated on held-out data and achieves a perplexity of approximately 19, a BLEU score of 0.40, a ROUGE-1 score of 0.21, and a ROUGE-L score of 0.19. These automatic metrics are used as baseline indicators, but the main evaluation emphasis is lexicography-oriented, meaning that the generated outputs are examined for usefulness in drafting and refining candidate example sentences, as well as for dictionary-critical errors such as hallucination, unidiomatic phrasing, register mismatch, and code-mixing leakage. The model is therefore positioned as an assistive tool for human-in-the-loop lexicographic curation rather than as a replacement for expert judgement. Overall, the study demonstrates how reproducible Sepedi corpus engineering can convert dispersed low resource text into reusable lexical resources and support practical computational lexicography.

Automated Annotation of Verbal Humor Based on a Multi-Agent System

欧阳晓芬 OUYANG Xiaofen

Abstract: The comprehension and generation of verbal humor constitute a core manifestation of advanced human language processing. Underlying this process are complex higher-order cognitive mechanisms, including semantic incongruity, contextual shift, and cultural schemata. For a long time, complex verbal humor in literary texts has largely depended on close reading and expert annotation. However, when applied to large-scale corpora, such methods face significant limitations, including low efficiency, strong subjectivity, and the difficulty of reproducing cognitive criteria. With the development of artificial intelligence, how to employ AI to simulate human cognitive processing of complex language has become a frontier issue at the intersection of linguistics and computational science.

Taking the identification of verbal humor in Qian Zhongshu's *Fortress Besieged* as a case study, this research constructs a multi-agent system that simulates human cognitive processing of humor. The system is grounded in the Semantic Script Theory of Humor (SSTH), the General Theory of Verbal Humor (GTVH), and cognitive-linguistic perspectives on frame shifting and irony. The study extracts 100 samples from *Fortress Besieged*, including 20 instances of explicit humor and 80 randomly selected textual passages, to form the experimental dataset. It then compares the performance of a single large language model with that of the multi-agent system in humor processing. Through the coordinated division of labor among agents responsible for preliminary screening, semantic script analysis, literary-context evaluation, rebuttal review, and arbitration, the multi-agent system successfully transforms the multi-level cognitive judgments involved in human humor processing into computable intelligent workflow procedures.

The experimental results show that the multi-agent system significantly outperforms the single model when dealing with complex humor that is highly dependent on irony,

degradation, and literary context. The system is capable of making accurate judgments about humorous attributes and generating intermediate explanatory chains concerning “script opposition” and “cognitive-schema integration,” thereby approximating the cognitive inferential process through which humans process verbal incongruity. This study demonstrates how traditional theories of cognitive linguistics can be transformed into computable and verifiable procedures for in-depth analysis, offering a new methodological pathway for the study of language cognition and processing in the digital-intelligent era.

数字人文视域下先秦神话人名识别与汉英词条编纂研究

秦曦 QIN Xi

摘要：先秦神话保留民族文化记忆，反映先民“中国梦”的初始形态。然而大多数先秦神话散佚于多部典籍中，且呈现出神话历史化或者神话哲学化的倾向。数字人文方法能够突破先秦文献的原有结构，提取人物、地点、事件等实体及实体间的关系，这为汉英词典人名词条的编纂提供了新视角。目前将数字人文的数据结果与双语词典编纂融合的研究较少。本文以《山海经》《淮南子》《庄子》的汉英数据为来源，通过智能体技术构建包含人物、地点、事件三类实体及其关联知识的聚合和可视化，构建先秦神话人名知识图谱，对比现有几部内向型汉英词典词条的文本分析结果，探讨智能体技术在汉英在线词典微观和中观结构设计的应用，以期拓展古籍数字化知识组织与智能化应用的研究与实践。

The Generation Mechanism of Crowdsourced Cultural Entries by EFL Learners in a Collaborative Intelligence Framework: A Case Study of the Spring Festival Entry

孙文龙 SUN Wenlong

Abstract: Rapid advancements in artificial intelligence are reshaping the landscapes of language learning and knowledge production. However, traditional cultural teaching often relegates learners to passive recipients of static information. From the perspective of collaborative intelligence, this study explores the generation process of crowdsourced cultural entries produced by EFL learners with the assistance of AI tools. The study takes English majors from a university in Nanjing as participants. Cultural descriptions of the Spring Festival written by volunteer students were collected, and the generation process of the cultural entry was analyzed through a case study based on the written materials and discussion records. The findings show that learners obtained cultural information from multiple sources, including AI tools and online resources, and gradually constructed a cultural knowledge framework through information selection and integration. In terms of human–AI interaction, the generation of the cultural entry demonstrates a collaborative pattern in which AI provides informational support while learners evaluate, reorganize, and refine the content. Regarding content organization and cultural expression, the generated entry shows two prominent features: a time-sequenced narrative describing the cultural process of the festival and an associative structure organized around key cultural concepts. The study suggests that cultural entry generation tasks in collaborative intelligent environments can enhance learners' active participation in cultural knowledge construction and improve their ability to express cultural content. Additionally, implications for cultural learning in foreign language education and for cross-cultural lexicography are also discussed.

Lexicographical Data Design for Tea-Related Terminology of GIAHS: A Perspective of Frame-based Terminology

王统 WANG Tong

Abstract: Driven by the growing demand for the international dissemination of Chinese tea culture and the knowledge translation of tea-related Globally Important Agricultural Heritage Systems (GIAHS), this study addresses a set of persistent inadequacies in existing dictionary articles of tea-related terminology: oversimplified definitions, insufficient representation of cultural connotations, unclear conceptual relations, and limited adaptability to diverse user needs. In response, the study draws on a lexicographic database approach and adopts Frame-based Terminology (FBT) as its theoretical framework to explore innovative data design and entry compilation for tea-related terminology.

The study proceeds in three stages. First, it delineates the scope and nature of tea culture terminology, and systematically evaluates the limitations of existing relevant dictionaries and terminological resources with respect to definitional adequacy, frame element representation, cultural information organization, relational network construction, and bilingual mediation support. Second, informed by the knowledge characteristics of tea culture and the practical needs of target users, the study identifies the core semantic frames and knowledge units underlying tea terminology. On this basis, it proposes a lexicographic database structure comprising seven components: the term entry, conceptual description, frame elements, cultural annotation, relational links, contextual exemplification, and translation/mediation information. The proposed model is then illustrated through a representative entry, jasmine tea (茉莉花茶).

The study is expected to make dual contributions: theoretically, it advances the integration of FBT with lexicographic database design; practically, it offers an operational and replicable model for tea terminology lexicography, the international communication of Chinese tea culture, and the knowledge translation of tea-related GIAHS.

The Applicability Boundary of General Large Language Models in Compiling Chinese Dictionaries for Foreign Learners of Chinese: A Case Study of DeepSeek-V4

王云霄 WANG Yunxiao

Abstract: General large language models (LLMs) have offered a promising new approach for enhancing both the quality and efficiency of compiling Chinese dictionaries for foreign learners of Chinese. However, existing studies tend to treat dictionary entries as homogeneous objects, and there is a notable lack of dedicated investigation into whether LLMs can reliably generate pattern information, which is one of the most distinctive features of learner’s dictionaries. Drawing on 250 in-syllabus words (productive-type) and 250 out-of-syllabus words (receptive-type) as research objects, this study employs DeepSeek-V4-Flash to generate dictionary entry information for both word categories, and conducts full-sample manual examination and quantitative analysis on the generated results. The findings reveal that the fully accurate rate is 39.6% for out-of-syllabus words but 0% for in-syllabus words, and that the total number of errors for in-syllabus words is 1.73 times that of out-of-syllabus words across six shared information dimensions. The generation of pattern information — a critical need for learners — proves to be a significant weakness of the model: the error rate of the pattern dimension for in-syllabus words reaches 96.85%, of which 75.28% are problems related to the internal structure of patterns, with errors concentrated in the complete representation of the five elements of Chinese patterns and the difficulty-graded ordering of collocational example words. The model performs reliably on objective-mapping tasks and demonstrates a distinctive strength in supplementing new senses and emerging usages; however, it underperforms on tasks requiring fine-grained judgment and professional normative tasks.

Based on these findings, this study delineates the applicability boundary of general LLMs in compiling Chinese dictionaries for foreign learners of Chinese along three dimensions: word category, information item, and task nature, and proposes a four-stage human-machine collaborative framework of “entry stratification – differentiated generation – tiered verification – iterative optimization”, in which LLMs are assigned to objective-mapping tasks at which they excel, while human compilers focus on fine-grained judgment tasks and professional normative tasks.

近五年新词语演变的多维透视

张静 ZHANG Jing

摘要：语言是社会的镜像，新词语的兴衰更替直接记录时代的脉动。本文以鲁东大学汉语辞书研究中心上线的《现代汉语融媒体词典》中收录的 2021-2025 年新词语为研究对象，从科技、民生、文化、治理四大领域切入，系统考察近五年中国社会的话语变迁轨迹。研究发现：科技类词语数量相较其他领域遥遥领先，2023 年 AI 大模型相关词语的集中爆发映射出人工智能技术的跨越式发展；民生类词语次之，从“三孩政策”到“育儿补贴”记录生育政策调整，从“就地过年”到“银发力量”反映社会结构变迁；文化类词语每年稳定在 28-32 条，“内卷”“躺平”“淡人/浓人”勾勒青年心态从焦虑表达到身份建构的演变图谱；治理类词语 94 条，呈现出逐年下降趋势，这完整记录了疫情防控话语从爆发到消退的周期。其他领域词语共计 256 条，涵盖经济产业、生态环境、国际关系、交通出行、体育健康等，反映社会生活的多元面向。研究表明，新词语的演变既是技术迭代的语言表征，也是社会心态的晴雨表，可以为理解当代中国社会转型提供独特的语言证据。

Beyond Etymology: Representing Motivational Information in English Learners' Dictionaries

赵连振 ZHAO Lianzhen

Abstract: Meaning is one of the core information categories in the microstructure of a dictionary entry. Traditional dictionaries, however, are primarily concerned with meaning description, that is, indicating what a headword means. Inspired by developments in cognitive linguistics, this study argues for a shift from meaning description to meaning explanation, namely, accounting for why a headword has a particular meaning. To this end, the term “motivational information” is proposed to designate the type of information that helps learners access the semantic motivations underlying lexical meaning. After examining deficiencies in existing lexicographical practice in English learners' dictionaries, the study argues for the need to move beyond purely etymological treatment towards motivational explanation and demonstrates the pedagogical value of such information for vocabulary acquisition and retention among EFL learners. It then establishes a typology of motivational information and, drawing on theories from cognitive semantics, proposes a framework for its systematic presentation in English learners' dictionaries. In light of the transformative impact of artificial intelligence on lexicography, the study further explores the potential of AI-assisted compilation of motivational information in English learners' dictionaries.